KNOWLEDGE CONTINUITY PLATFORM · HACKATHON ROUND 1

# Continuity, when people move on

## Prevent Critical Knowledge From Leaving With People

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THE PROBLEM

# When people leave, knowledge leaves with them

Project truth is distributed across documents, conversations, code, task boards and meeting notes.

The incoming member receives fragments, not context.

**Official challenge:** “Build a system that converts scattered project activity into a usable handover when a team member leaves.”



WHY IT MATTERS

# Knowledge loss creates a continuity tax

When context disappears, the project pays in time, confidence and repeated effort.

01

## Fragmentation

Facts sit in separate tools.

02

## Uncertainty

No clear picture of what matters.

03

## Rework

People repeat questions and mistakes.

04

## Delay

Decisions and ownership drift.

**Cause → fragmentation → uncertainty → rework → delay**




THE USERS

# Everyone feels the gap differently

## Departing member

Cannot transfer tacit context efficiently.

## Incoming member

Does not know what matters first.

## Project manager

Lacks a reliable continuity view.

## Developer

Inherits code without the decision trail.

## HR / administrator

Needs a safe transition process.

## Students, researchers and volunteers

Rotate frequently.

MVP focus: student project teams and small technical teams, where source diversity is real but permissions and scope remain manageable.

THE INSIGHT

# Documentation is not the same as knowledge

A record of activity is not yet an explanation of project state.

- What happened is not the same as why it happened.
- A closed task can hide an unfinished dependency.
- A decision matters because it changed the project.
- A useful handover explains what remains, what is blocked and what happens next.



**Our insight:** continuity requires reconstructing context and action, not merely summarising files.

THE PROPOSED PRODUCT

# Meet ContinuityOS

An evidence-backed project knowledge reconstruction and continuity platform.

It turns authorised project activity into a reviewable handover for the next person responsible.

## Value

Faster understanding without guessing.

## Audience

Incoming members, project leads and administrators.

Proposed product concept — not yet implemented.



THE CORE LOOP

# From scattered activity to a trusted handover

1. **Connect** — authorised sources in; scoped project data out
2. **Reconstruct** — decisions, ownership, risks and dependencies
3. **Verify** — evidence links, gaps, conflicts and uncertainty
4. **Review** — authorised humans correct and approve
5. **Handover** — role-appropriate continuation plan

**Each stage has an explicit input, operation and reviewable output.**

THE EXPERIENCE

# The incoming member receives a map, not a data dump

The handover is organised around decisions, ownership and next actions.

- Project overview and current state
- Decision timeline with evidence references
- Completed and pending work
- Ownership, dependencies, blockers and risks
- Unresolved questions and recommended next actions
- Source links beside important claims

Conceptual dashboard mockup — proposed experience.

CURRENT STATE

## Handover overview

What changed, what remains, and where attention goes next.

EVIDENCE

## Decision trail

Claims linked to authorised source records.

ACTION

## Next actions

Owners, blockers and unresolved questions.

THE DIFFERENTIATOR

# Beyond summarisation.
# Towards knowledge reconstruction.

## Evidence-first intelligence

Claims point back to authorised sources.

## Context reconstruction

Discussions, decisions and actual changes connect.

## Knowledge-gap detection

Missing, conflicting and uncertain information surfaces.

## Dependency-aware handover

Blocked work and next actions become visible.

**Proposed completeness score:** coverage of decisions, owners, pending work, risks, dependencies and next actions, with evidence quality and unresolved gaps visible, not hidden.

# The intelligence pipeline

Authorised sources

Ingestion.

Extraction

Document, code and task extraction → normalisation.

Semantic indexing

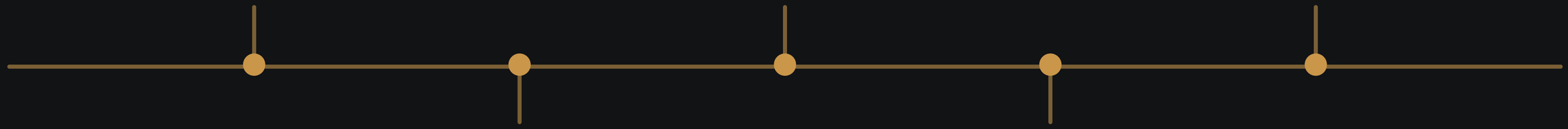
Relationship identification.

Project-state reconstruction

Evidence validation.

Handover generation

Human review.



**LLMs**
Draft structured facts.

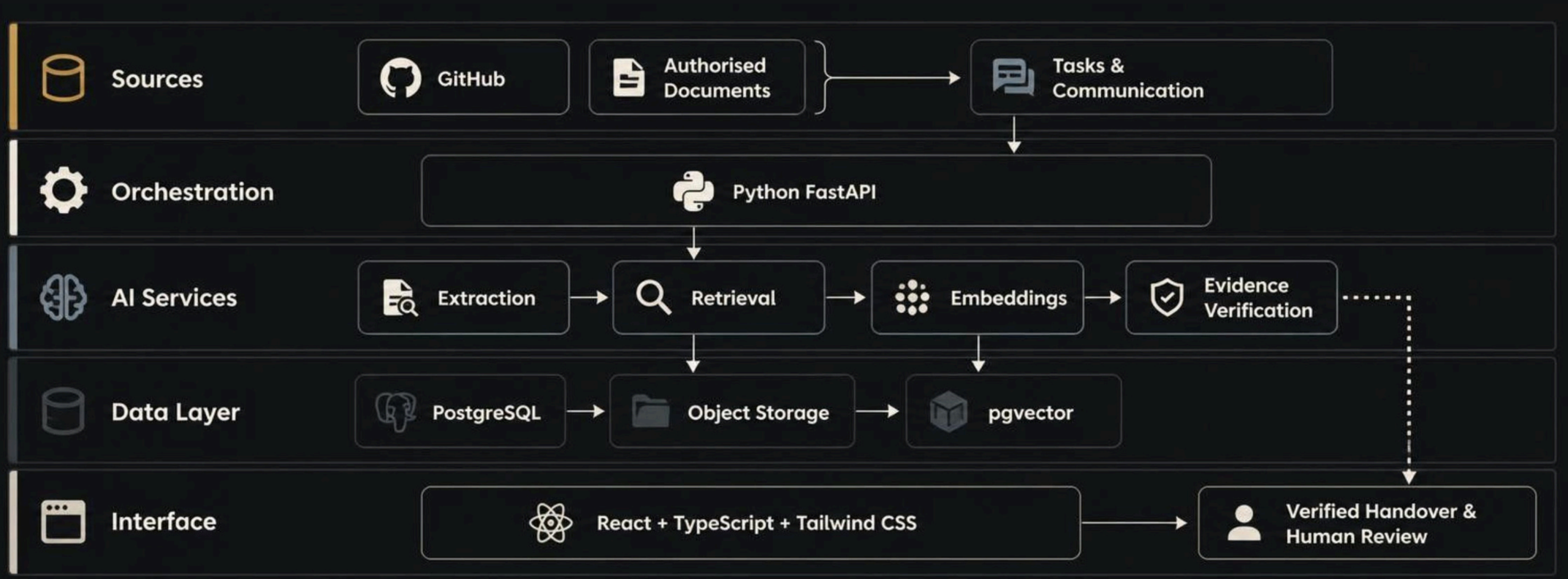
**Embeddings**
Support retrieval.

**Deterministic checks**
Preserve provenance.

Missing information stays missing.

BUILD PLAN

# A modular architecture we can build




Flow: sources → FastAPI → extraction / retrieval → Postgres + pgvector → verified handover → React review.

TRUST BY DESIGN

# Trust is a product requirement



## Access

Authentication and role-based access control.



## Scope

Project-level and source-level permission checks.



## Protection

Sensitive-data redaction and secure storage.



## Audit

Logs for ingestion, edits, approvals and sharing.



## Review

Human approval before a handover is shared.

**Conflicting claims appear as conflicts. Missing evidence appears as a gap. Unauthorised content never enters the handover.**

MVP SCOPE

# Scope the MVP around the handover moment

## MVP

Project creation, document and repository ingestion, extraction, timeline, evidence-linked summaries, ownership, risks, dependencies, review and handover generation.

## Advanced

Multiple integrations, dependency graph, semantic search, conflict detection and completeness assessment.

## Future

Continuous monitoring, cross-project discovery, automated onboarding and multilingual reconstruction.

**Round 1 proves the plan; Round 2 proves the core loop.**

TECHNICAL FIT

# Why this stack fits the challenge

### React + TypeScript

Fast, maintainable review interface.

### Tailwind CSS

Consistent enterprise UI language.

### Python + FastAPI

Clear AI and integration orchestration.

### Supabase / PostgreSQL

Relational project state, auth and storage.

### pgvector

Pragmatic semantic retrieval when required.

### LLM + embeddings

Structured extraction and context-aware generation.

**GitHub API + Vercel / Render:** Feasible first integration and deployment.

EXECUTION

# One team, one continuity loop

## [NAME] — Product architecture & leadership

Scope, system design, judge narrative.

## [NAME] — Frontend & UI/UX

Dashboard, review flows, accessibility.

## [NAME] — Backend & APIs

Ingestion, orchestration, permissions.

## [NAME] — AI/ML engineering

Extraction, retrieval, evidence checks.

## [NAME] — Data & integrations

Schema, GitHub and source connectors.

## [NAME] — QA & documentation

Test plan, demo readiness, submission quality.

Replace placeholders with the team’s official names and actual ownership before submission.

DELIVERY PLAN

# Three rounds, three kinds of proof

ROUND 1

## The plan

Problem analysis, users, architecture, prioritisation and feasibility.

Only this work is represented here as completed.

ROUND 2

## The build

Frontend, source ingestion, AI processing, evidence-linked handover, review and approval, working MVP demo.

ROUND 3

## The finish

Reliability, evidence verification, dependency intelligence, permission enforcement, UX and deployment.

ROUND 2 ACCEPTANCE TESTS

# Three commitments we will earn after Round 2

## 01 Evidence reliability and contradiction detection

Measure supported-claim coverage and show conflicts instead of merging them.

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## 02 Dependency graph and continuation planning

Demonstrate blocked-task paths and next actions.

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## 03 Role-aware permissions and personalisation

Show different approved handovers for different authorised roles.

Each improvement is demonstrated with a before/after test scenario, not a promise without a check.

EVALUATION

# Impact we will measure, not invent

## Individual

Time to understand ownership, risks and next actions.

## Team

Fewer repeated questions, clearer accountability and faster transition.

## Organisation

Preserved context and reduced transition overhead.

**Evaluation plan:** compare a source-only handover with the proposed evidence-backed flow using task completion time, supported-claim rate, unresolved-gap visibility and reviewer correction rate.

Results will be reported after testing.



THE LONGER HORIZON

# From handover tool to project memory

## Continuity

Continuous project memory and more workplace integrations.

## Analytics

Multilingual reconstruction and organisational knowledge analytics.




## Enterprise intelligence

Enterprise knowledge continuity and cross-project intelligence.

## Onboarding

Automated onboarding and advanced relationship mapping.

Potential adopters: startups, enterprises, universities, research groups and NGOs.

# People may leave projects. Their knowledge shouldn’t.

ContinuityOS · Evidence-backed handovers for projects in motion.

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Thank you.

